ECE448 Spring 2026

L07-am-simulation

UNITED STATES
AIR FORCE
ACADEMY

Today

- Sampling rate demo
- Data types demo
- Filtering
- AM simulation
- AM receiver

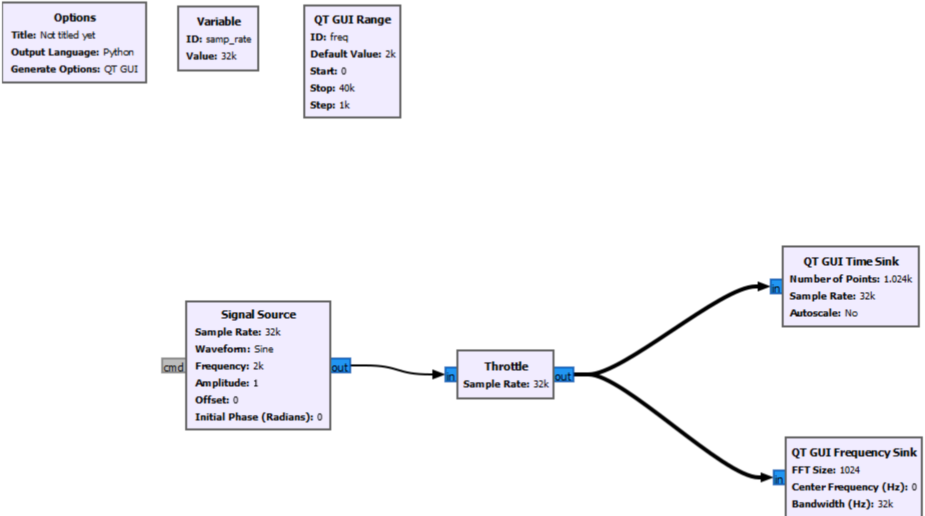
SAMPLING RATE

- Remember DSP? How do we measure where aliases end up?

- With f_s = 32ksps, predict where these frequencies will end up:

f_0	**Alias Prediction**	**Alias Measured**
2kHz		
14kHz		
16kHz		
18kHz		
24kHz		

Sampling Rate

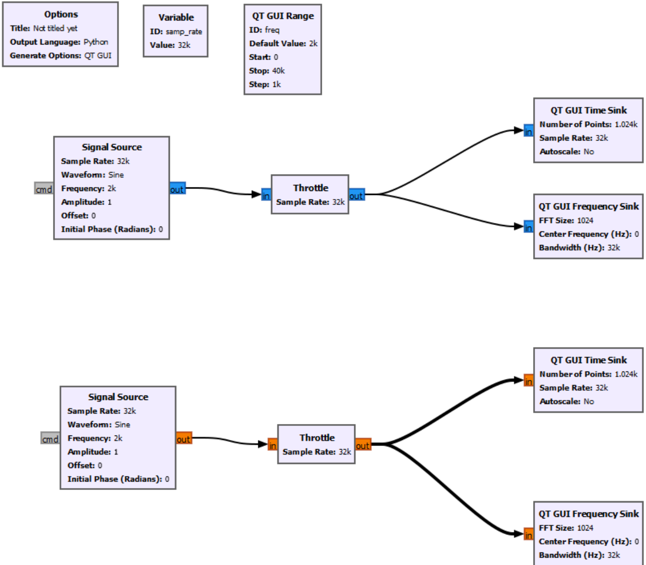


Data Types

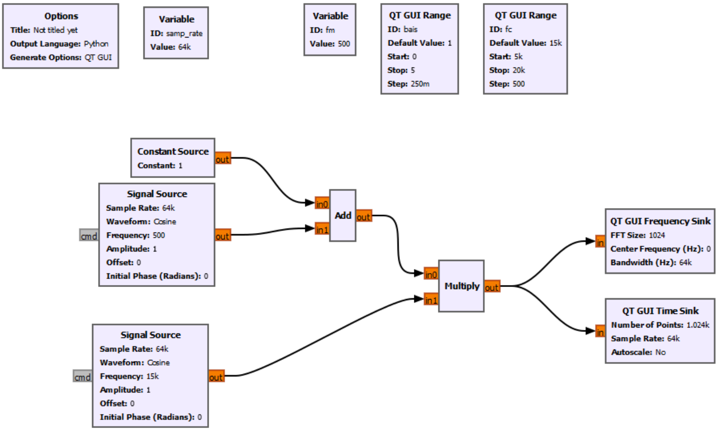
- Float $\longrightarrow$ ________________
- Complex $\longrightarrow$ ________________
- Check in python...

```
import numpy as np
import matplotlib.pypl
f = np.fromfile(open('
plt.plot(f)
plt.show()
```

Data Types


Options
Title: Not titled yet
Output Language: Python
Generate Options: QT GUI
Variable
ID: samp_rate
Value: 32k
QT GUI Range
ID: freq
Default Value: 2k
Start: 0
Stop: 40k
Step: 1k
Signal Source
Sample Rate: 32k
Waveform: Sine
Frequency: 2k
Amplitude: 1
Offset: 0
Initial Phase (Radians): 0
cmd
out
in
Throttle
Sample Rate: 32k
out
in
QT GUI Time Sink
Number of Points: 1.024k
Sample Rate: 32k
Autoscale: No
in
QT GUI Frequency Sink
FFT Size: 1024
Center Frequency (Hz): 0
Bandwidth (Hz): 32k
Signal Source
Sample Rate: 32k
Waveform: Sine
Frequency: 2k
Amplitude: 1
Offset: 0
Initial Phase (Radians): 0
cmd
out
in
Throttle
Sample Rate: 32k
out
in
QT GUI Time Sink
Number of Points: 1.024k
Sample Rate: 32k
Autoscale: No
in
QT GUI Frequency Sink
FFT Size: 1024
Center Frequency (Hz): 0
Bandwidth (Hz): 32k

Linear Modulation



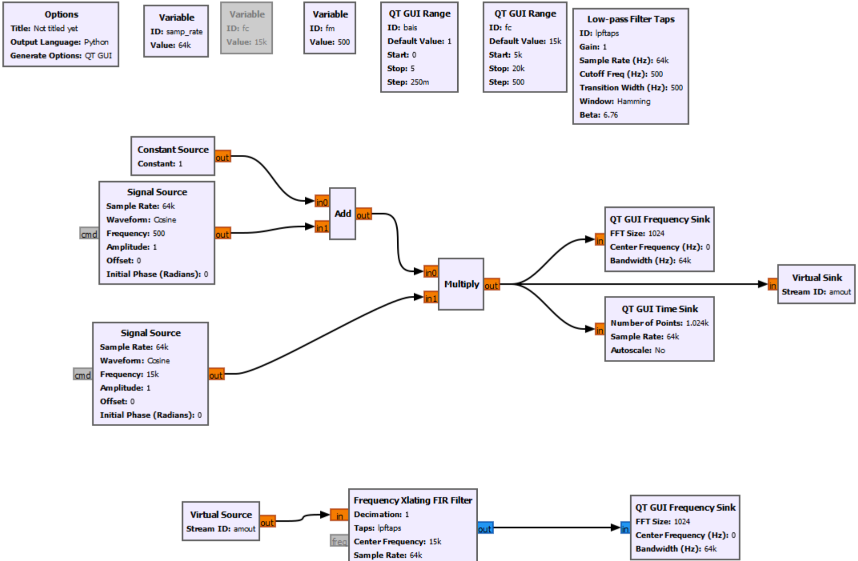
Filters

- https://pysdr.org/content/filters.html

- Output of a filter is ________________ of filter's impulse response and the ________________

- Good news...

FILTERS


Options
Title: Not titled yet
Output Language: Python
Generate Options: QT GUI
Variable
ID: samp_rate
Value: 64k
Variable
ID: fc
Value: 15k
Variable
ID: fm
Value: 500
QT GUI Range
ID: bais
Default Value: 1
Start: 0
Stop: 5
Step: 250m
QT GUI Range
ID: fc
Default Value: 15k
Start: 5k
Stop: 20k
Step: 500
Low-pass Filter Taps
ID: lpftaps
Gain: 1
Sample Rate (Hz): 64k
Cutoff Freq (Hz): 500
Transition Width (Hz): 500
Window: Hamming
Beta: 6.76
Constant Source
Constant: 1
out
Signal Source
Sample Rate: 64k
Waveform: Cosine
Frequency: 500
Amplitude: 1
Offset: 0
Initial Phase (Radians): 0
cmd
out
in0
in1
Add
out
in0
in1
Multiply
out
QT GUI Frequency Sink
FFT Size: 1024
Center Frequency (Hz): 0
Bandwidth (Hz): 64k
in
Virtual Sink
Stream ID: amout
QT GUI Time Sink
Number of Points: 1.024k
Sample Rate: 64k
Autoscale: No
Signal Source
Sample Rate: 64k
Waveform: Cosine
Frequency: 15k
Amplitude: 1
Offset: 0
Initial Phase (Radians): 0
Virtual Source
Stream ID: amout
Frequency Xlating FIR Filter
Decimation: 1
Taps: lpftaps
Center Frequency: 15k
Sample Rate: 64k
freq
QT GUI Frequency Sink
FFT Size: 1024
Center Frequency (Hz): 0
Bandwidth (Hz): 64k

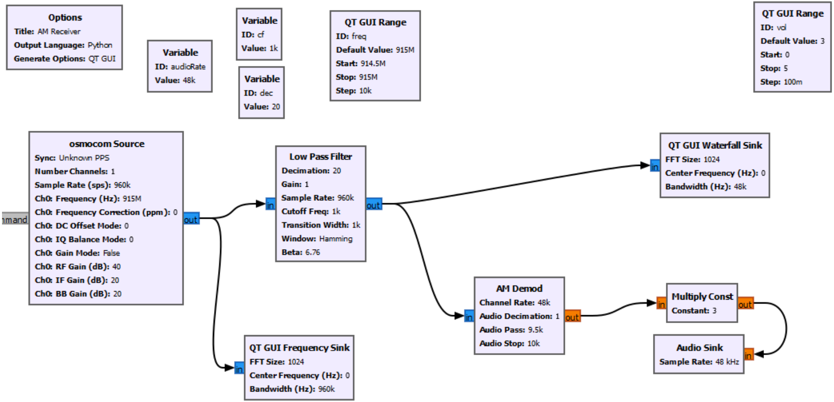
Linear Demodulation

Your turn

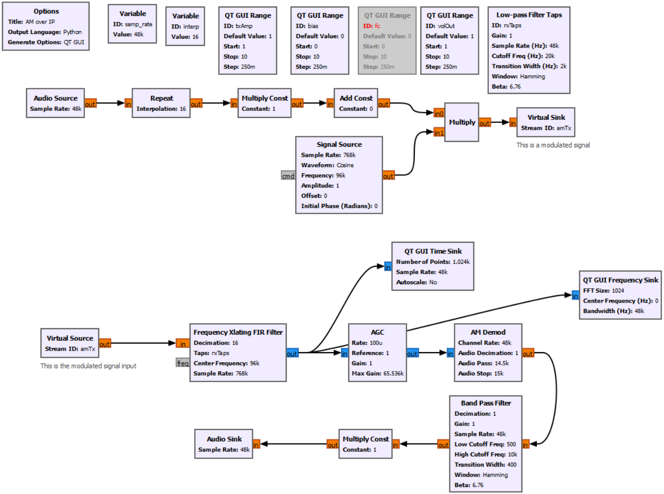
Create an AM receiver/demodulator...

- What elements do you need?

Linear Demodulation



AM: Full Signal Chain



SSB Modulation

Homework

- Create an AM receiver and record audio (recommend ATIS at 128.525 MHz)
- Create a phase shift SSB modulator using the Hilbert transform.
- Turn in: *.GRC files & *.wav file for AM audio